



Thermofluidic characteristic of a nanofluid-cooled oblique fin heat sink: An experimental and numerical investigation

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ARTICLE INFO

Keywords:

Oblique fin heat sink
Al₂O₃-Water nanofluids
RNG k-ε Model
Volumetric concentration
Performance factor

ABSTRACT

This paper presents an experimental and numerical investigation on the thermofluidic performance of 0.5–2.0 % of the volumetric concentration of Al₂O₃-water nanofluids through oblique fin heat sink microchannel (OFHS MC) in the Reynolds number range 100–300. In comparison, experiments were also performed using a straight channel heat sink microchannel (SCHS MC) of the same hydraulic diameter as the oblique fin channel. The fluid flow is simulated using the RNG k-ε model in both periodic and full domain numerical analysis. The phenomenological behaviour of the nanofluid flow is further studied using numerical results. The oblique fin microchannel encouraged the secondary flow, thus enhanced the macroscopic mixing. Boundary layer disruption and vortices generation in the secondary channel is the critical phenomena of improved heat transfer. A substantial improvement of heat transfer coefficient is achieved for nanofluid and oblique fin heat sink combinations compared to the straight channel heat sink. The heat transfer coefficient is increased at increasing nanofluid concentrations for any given Reynolds number and channel configuration. The enhancement of about 55 % in the heat transfer coefficient with a 2 % volumetric concentration of nanofluid is achieved at Re = 300 in an oblique fin heat sink. The heat transfer enhancement is always significant than the pressure penalty. The oblique fin heat sink improves the average junction temperature considerably compared to the straight channel heat sink. The inclusion of nanofluid can reduce the experimental junction temperature by up to 6 °C. Therefore, the combined advantage of the oblique fin heat sink with nanofluid is anticipated as one of the potential alternatives for electronics cooling.

1. Introduction

Thermal management is a critical part of keeping all effective electronics systems working and robust. Microelectronic mechanical systems allow innovative cooling techniques to be used in miniature devices, particularly in electronics, as the world moves towards smaller sizes, cheaper materials, lighter weights, and greater performance. Improving the electronic chip's performance involves more resistors, which will cause significant heat generation in these devices. This obliges researchers to build a cooling system, which is more efficient. One of the most promising cooling solutions for microelectronic devices has become the microchannel heat sink developed by Tuckerman and Pease [1]. They performed a pioneer work in 1981, established the concept of microchannel heat sinks, and predicted the superior heat flux removal rate up to 1000 W/cm². Their invention of microchannels played a vital role in the heat transfer phenomena in the later decades. Kandlikar et al. [2] showed detailed attention towards the single-phase

gas and fluid flow, boiling, condensation, single-phase electro-kinetic flow, and microchannel potential in the biomedical field.

The cooling capacity of microchannel exhibits its limit with conventional coolant (water); therefore, there is a need for a new class of fluid called nanofluid with the novel microchannel. Choi and Eastman [3] acquainted “nanofluids” contrived by nanoparticle suspension in the base fluids. The tiny nanoparticles and a base fluid combine to form nanofluids [4]. Compared with conventional coolants, nanofluids show tremendous potential for heat transfer improvement. Thereby, non-scientists and engineers around the world have drawn enormous attention to nanofluids. The heat transfer improvement approaches, using nanofluid as a coolant, have been demonstrated by various researchers [5,6]. So far, several research studies have shown an increase in heat transfer for different configurations of microchannels with nanofluids.

Firstly, Jang and Choi [7] studied the heat transfer enhancement in the microchannel with water-based nanofluid. The study revealed the

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<https://doi.org/10.1016/j.ijthermalsci.2021.107214>

Received 25 May 2021; Received in revised form 27 July 2021; Accepted 4 August 2021

Available online 13 August 2021

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higher dissipation of heat flux in the range of 1300–2000W/cm² by the nanofluid cooled microchannel heat sink. Huang et al. [8] conducted the analytical study of nanofluids in a 3D microchannel heat sink. They observed significant heat transfer and no additional pressure drop because the particle size is small and concentration is low. Chein and Chuan [9] experimentally investigated CuO/water nanofluid performance in a microchannel heat sink. Their result revealed the enhancement of heat transfer compared to water at a lower flow rate, while at a higher flow rate, no enhancement is achieved. In their numerical studies, Mohammed et al. [10] found that at a lower concentration (<5 %), nanofluid is capable of increasing the heat transfer, while at 5 % concentration, it is unable to enhance the further heat transfer. With different nanoparticles (Al₂O₃, ZnO, SiO₂, and CuO), Kuppusamy et al. [11] contemplated grooved trapezoidal geometry and investigated the influence of parametric design and found that nanoparticle (25 nm-diameter) maximizes heat transfer at 4 % concentration. Hasan [12] studied three types of pin fins, i.e., triangular, square, and circular fin with diamond and Al₂O₃ nanoparticles based nanofluid and found that nanofluid significantly enhances the heat transfer with higher pressure drop. In addition, the circular pin fin having the highest heat transfer rate among all. Duangthongsuk and Wongwise [13] compared the thermo-hydraulic performance in the various heat sinks (square and miniature circular pin fin structure) with nanofluid (SiO₂ dispersed in the DI water) at 0.2, 0.4 and 0.6% of the particle volume fraction. The heat transfer rate of SiO₂-water nanofluid was observed 4–14 % higher than water. Further, they studied the various channel structures zigzag flow with different geometry, such as cross-cutting and continuous [14]. The geometry with cross-cutting showed a 2–6% higher enhancement than continuous geometry. In addition, nanofluid-cooled geometry delivered a 3–15 % higher enhancement than base fluid.

Mohammadpour et al. [15] used the Discrete phase model to simulate nanofluids in a microchannel with a synthetic jet. They discovered that, by reducing particle size, increasing particle concentrations, and boosting membrane amplitudes and frequencies, enhancement in heat transfer had been observed. Singh et al. [16] conducted a detailed experimental and numerical analysis for the convective heat transfer research in the microchannel with nanofluid. They have also shown that Al₂O₃-water nanofluids are not usually associated with changes in structures, and some other approaches, including dispersion, result in enhanced heat transfer. Ahmadi et al. [17] used the finite volume approach to simulate a square channel with a cylinder in the middle connected to wavy fins at varying concentrations of an Al₂O₃ nanofluid. By raising the concentration from 1 % to 3 %, the heat transfer coefficient improves by just 8 %, but the pumping power increases by around 60 %.

To improve the performance of the heat sink, Khoshvaght-Aliabadi et al. [18,19] investigated the presence of Al₂O₃-water nanofluid on the straight and oblique pin fin geometry. They found that the augmentation of heat transfer is better in nanofluid-transported heat sink compared to base fluid transported heat sink. Feng et al. [20] investigated the influence of average thermofluidic characteristics in an interrupted microchannel with double circular pin-fins (IMCDCP) in each microchamber at Reynolds numbers ranging from 133 to 550. The proposed IMCDCP may increase overall thermal performance while also achieving low wall temperature and good wall temperature uniformity.

Wang et al. [21] numerically and experimentally investigated microchannel heat sink (MCHS) with grooves and micro-ribs. They found that the MCHS has led to a high-pressure drop penalty. Bahiraei et al. [22] examined nanofluid in a microchannel heat sink having ribs and secondary channels. The results show that employing the nanofluid in the microchannel heat sink enhanced with the ribs and secondary channels significantly improves heat sink performance. The flow experienced higher pumping power at higher Reynolds numbers and concentrations. Bahmani et al. [23] numerically studied the Al₂O₃-water nanofluid in a counter flow heat exchanger. They observed the maximum thermal efficiency of a heat exchanger at 5 % of volumetric

concentration. The further increase of nanoparticle concentration leads to a reduction in thermal efficiency. The Nusselt number and thermal efficiency were observed at 32.7% and 30%, respectively. Akbari et al. [24] employed the nanofluid in a 3-D curved microtube to investigate heat transfer. The results showed that the thermofluidic behaviour of nanofluids in curved geometries is different from other geometries and flat surfaces due to buoyancy and centrifugal forces at the same time. Tian et al. [25] studied the thermofluidic characteristics in a spiral double-tube heat exchanger. The high heat transfer coefficient was observed due to the conversion of fluid motion into secondary motion.

Extensive research has shown that utilizing nanosuspensions (NS) provides several advantages and improvements over traditional fluids [26–29]. Ho et al. [26] used diverging tubes and nano-PCM/water NS (nanosuspension). The results revealed that utilizing NS reduces thermal resistance by around 5 % compared to the base fluid. Ho et al. [27] studied the thermal-hydraulic behaviour of an Al₂O₃/water nanofluid in a minichannel heat sink. They determined that the Nusselt number was temperature-sensitive and could be reduced by employing an Al₂O₃-water nanofluid at higher inlet temperatures. Yan et al. [29] quantitatively investigated convective heat transport of Al₂O₃-water nanofluid in a minichannel heat sink with a micro-encapsulated PCM-cooled ceiling. The results indicate that the thermal resistance lowers around 10.88% at a solid volume fraction of 10 % at Re_{bf} = 500 and T_{cw,0} = 28 °C for the case of the bare ceiling. Sarafraz et al. [30] demonstrated experimentally that the use of nano-platelets nanosuspensions might potentially increase the working fluid thermal conductivity by 32.1 %. Goodzari et al. [31] conducted a numerical investigation in a microchannel with several cross-sections of heat transfer of graphene-silver/water hybrid nanofluids with a volume fraction of 0–0.1 vol percent. The authors observed that heat transmission increased significantly with the concentration of nanofluids.

Headed towards heat transfer enhancement, the heat sink acts as a critical factor because of the convective resistance contribution to total thermal resistance [32]. For lowering the convective resistance, great exertion has been made towards the upsurge in the heat sink design. The conventional heat sink attains its limit and is incompetent to comply with the present-day demand of removing the heat flux with a lesser pressure penalty. Therefore, several novel ideas or passive methods have been come out to replace the conventional configuration. Several factors, such as secondary flow for boundary layer reinitialization or increasing the mixing of flow through regular or chaotic advection, mainly capitalized the flow phenomena. The secondary flow is one of the promising options for better fluid mixing, and disruption of the boundary layer promotes heat transfer enhancement. Kandlikar [33] suggested two types of methods for the generation of secondary flow (1) a smaller channel is introduced at some angle between two main channels and (2) provide the constriction area towards the larger section so that the venturi effect helps to generate secondary flow. The secondary flow utilization for different heat sink configuration have been conducted by several researchers in their study ([18,19,34,35], and [36]).

Lee et al. [37] first introduced oblique fins in the microchannel heat sink. A substantial enhancement in heat transfer (almost double) was recorded using oblique cuts on straight channels. Further, the study on air-cooling oblique fin [38,39], liquid cooling (single-phase and two-phase boiling) [40–42] has been extensively conducted and reported as the best performing design among many designs [43]. The present study authors [44,45] have also reported the oblique-finned heat sink's nanofluid cooling performance through numerical analysis.

The combined effect of the oblique fin and the nanofluids on improving the thermal performance within the oblique fin heat sink is worth anticipating, based on the above research results. However, this may accompany a complicated mechanism, especially for nanofluids. Therefore, the applicability of nanofluids to the oblique fin should also be further explored. Besides, it can be found from the literature survey that the experimental studies incorporating nanofluids in oblique

geometry may not have been taken into account, and this has inspired the present research. We close this gap by studying 3D conjugate heat transfer in nanofluid-cooled OFHS. The current experimental investigation explores the thermofluidic characteristic of an oblique fin heat sink with Al_2O_3 -water nanofluid at 0.5, 1, 1.5 and 2 % vol%. Besides the effect of nanoparticle volume concentration on enhancing the heat transfer characteristics, the thermofluidic characteristic in terms of the flow and temperature fields from the numerical results has been demonstrated. This OFHS is anticipated to be an excellent alternative for electronics cooling applications.

2. Description of model design and experimental methodology

The straight channel heat sink (SCHS) and the oblique fin heat sink (OFHS) test section have been presented in Fig. 1. The footprint ($25 \text{ mm} \times 25 \text{ mm}$), channel height (1.5 mm), and channel width (0.5 mm) of both models are the same, as shown in Fig. 1. The height of the channel (H_{ch}) is 1.5 mm, the width of the channel (W_{ch}) is 0.5 mm, the width of the fin (W_{fin}) is 0.5 mm, and the width of the heat sink (W_{HS}) is 25 mm. Based on the conventional SCHS footprint, the OFHS is created. The OFHS has been obtained due to secondary passage in SCHS fin in the collinear pattern. The oblique angle (θ) is 27° . The secondary width (W_{ob}) of the channel is 0.23 mm. The pitch of the fin (P_{fin}) is 2 mm. The other dimension of the OFHS is similar to SCHS. The sectional fins interrupted boundary layers and maintained flow in the developing zone. The secondary flow is introduced by the oblique channels, improving flow mixing. The boundary layer reinitialization and flow mixing could significantly improve the heat transfer performance by maintaining the flow in a continuous developing state and thus, improving advection. SOLIDWORKS [46] developed the prototypes of the heat sinks and processing performed by CNC milling. The extended part known as the flange is given to fix the test section (heat sinks) in the housing to make it leak-proof before experimenting. The bottom portion of the heat sink contained four holes for cartridge heaters. The details regarding this will be discussed in the subsequent section.

3. Computational investigation and analysis

3.1. Assumptions

In the present investigation, the following assumption has been considered to conduct the numerical analysis.

- No-slip is considered between the nanoparticle and water (base fluid).
- The uniform dispersion of solid nanoparticles and the size of the nanoparticle are deemed ultrafine.
- The fluid is incompressible and single phase.
- The heat transfer is three-dimensional and steady.
- The following conditions are ignored: compressibility effect, radiation heat transfer, viscous dissipation and gravitational force.

3.2. Governing equation implemented

The computational analysis of water/ Al_2O_3 -water nanofluid through oblique fin needs to carry out with the turbulence model to get satisfactory convergence for all flow rates due to the structure of oblique fins. The description of fluid flow phenomena seen in the various complex structure was summarized [47]. As a result, the RNG k- ϵ model (a refined version of the k- ϵ model) [48] is employed. The RNG k- ϵ model has enough potential or performs better to analyze the rapidly strained flow inside the model. The traditional k- ϵ model is generally a high Re model. Simultaneously, the RNG theory offers an analytically derived differential formula for effective viscosity that accounts for low Reynolds number effects. However, the proper treatment of the near-wall region is needed for the effective use of this function. These features improve the accuracy and applicability of the RNG k- ϵ model for the present study. They are more accurate than the traditional k- ϵ model for a broader range of flows. Using a mathematical technique known as “renormalization group” (RNG) methods, the RNG-based k- ϵ turbulence model is derived from the instantaneous Navier-Stokes equations. Chen et al. [49] showed that the RNG k-turbulence model predicted the performance of fin and tube heat exchangers more precisely than the

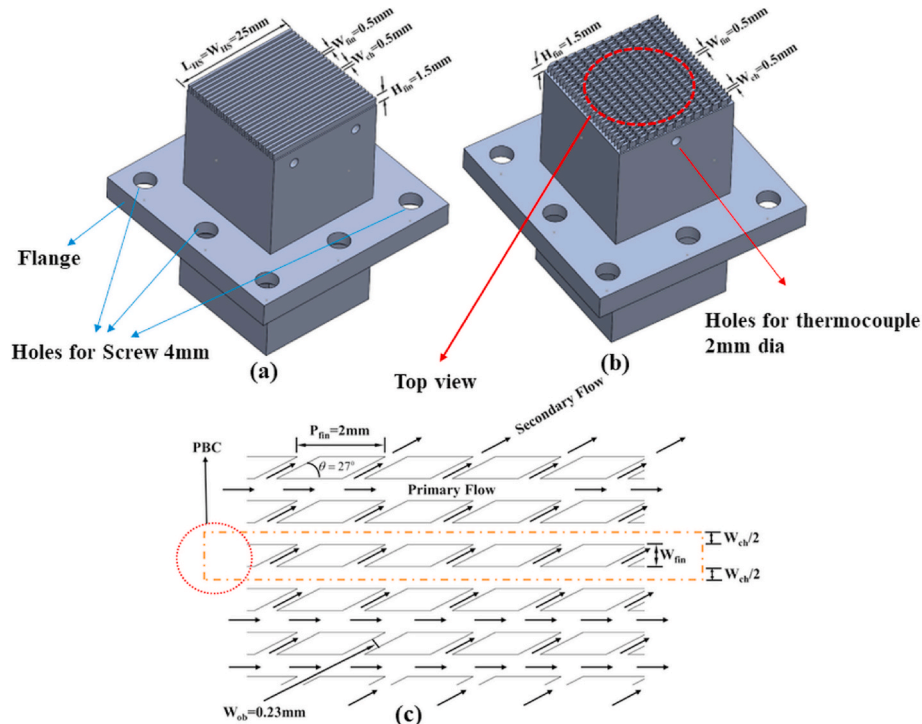


Fig. 1. 3D Schematic isometric view of (a) SCHS (b) OFHS (c) Top view of OFHS.

laminar model at low flow rates. This model's accuracy in resolving flow and temperature fields of related complex heat sink fin designs has already been demonstrated in Refs. [50,51]. The governing equations involved in the numerical investigation is shown below:

$$\frac{\partial}{\partial x_i}(\rho v_i) = 0 \quad (1)$$

$$\frac{\partial}{\partial x_j}(\rho v_j v_j) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) - \rho \overline{v_i' v_j'} \right] \quad (2)$$

$$\frac{\partial}{\partial x_j}(\rho v_j T) = \frac{\partial}{\partial x_j} \left[\frac{\mu}{Pr} \frac{\partial T}{\partial x_j} - \rho \overline{T' v_j'} \right] \quad (3)$$

$$\frac{\partial}{\partial x_j} \left(k \frac{\partial T}{\partial x_j} \right) = 0 \quad (4)$$

The term $-\rho \overline{v_i' v_j'}$ and $-\rho \overline{T' v_j'}$ is called turbulent stress and turbulent heat flux respectively and defined as;

$$-\rho \overline{v_i' v_j'} = \mu_t \left(\frac{\partial v_j}{\partial x_i} + \frac{\partial v_i}{\partial x_j} \right) - \frac{2}{3} \left(\rho k + \frac{\partial v_k}{\partial x_k} \mu_t \right) \delta_{ij} \quad (5)$$

$$-\rho \overline{T' v_j'} = \frac{\mu_t}{Pr_t} \left(\frac{\partial T}{\partial x_j} \right) \quad (6)$$

The following transport equations yield the turbulence kinetic energy, k , and its rate of dissipation, ϵ . After considering all of the current investigation assumptions, an updated/modified version of the RNG $k-\epsilon$ model is presented compared to the regular $k-\epsilon$ model.

$$\frac{\partial}{\partial x_i}(\rho k v_i) = \frac{\partial}{\partial x_j} \left(\alpha_k \mu_{eff} \frac{\partial k}{\partial x_j} \right) + G_k - \rho \epsilon + S_k \quad (7)$$

$$\frac{\partial}{\partial x_i}(\rho \epsilon v_i) = \frac{\partial}{\partial x_j} \left(\alpha_\epsilon \mu_{eff} \frac{\partial \epsilon}{\partial x_j} \right) + C_{1\epsilon} \frac{\epsilon}{k} G_k - C_{2\epsilon} \rho \frac{\epsilon^2}{k} \quad (8)$$

$$G_k = \mu_t \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) \frac{\partial v_i}{\partial x_j} \quad (9)$$

S_k is a user-defined term; the term G_k indicated the generation of

turbulence kinetic energy due to mean velocity gradients. The term μ_{eff} and μ_t are the viscosities (effective and turbulent (eddy)) and are presented in (10) and (11), respectively.

$$\mu_{eff} = \mu + \mu_t \quad (10)$$

$$\mu_t = \frac{\rho C_\mu k^2}{\epsilon} \quad (11)$$

The term $C_{2\epsilon}^*$ is a constant (in the modified RNG $k-\epsilon$ model).

$$C_{2\epsilon}^* = C_{2\epsilon} + \frac{C_\mu \eta^3 \left(1 - \frac{\eta}{\eta_o} \right)}{1 + \beta \eta^3} \quad (12)$$

where, $\eta = \frac{S_k}{\epsilon}$ the model constant (analytically derived) from the RNG theory is $C_\mu = 0.0845 C_{1\epsilon} = 1.42 C_{2\epsilon} = 1.68 \eta_o = 4.38 \beta = 0.012, \alpha_k = \alpha_\epsilon = 1.393$ [52].

3.3. Model setup and boundary conditions

The thermofluidic features of SCHS/OFHS with enhanced grid resolution were investigated using a single domain model with periodic boundary conditions (PBC), as shown in Fig. 2(a). In addition, as shown in Fig. 2(b), a full domain simulation was also performed. The water/nanofluid temperature is 298.15 K at the entrance of the fluid domain in both numerical and experimental conditions. The heat sink inlet and outlet of the model were extended 40 mm to match the testing needs. The wetted wall of the heat sink is coupled with the wetted wall of the fluid (water/nanofluid) domain with a no-slip boundary condition. The uniform heat flux 64 W/cm² is given at the heat sink's bottom footprint (base/junction). The top, bottom, and sidewall of the extended portion of the fluid domain are adiabatic. Except for the base of the heat sink, all outer boundary surfaces are considered to be adiabatic. At the outlet, a pressure outlet boundary condition has been applied. The full domain model's boundary condition (at the inlet and outlet) is similar to the periodic model used in this study.

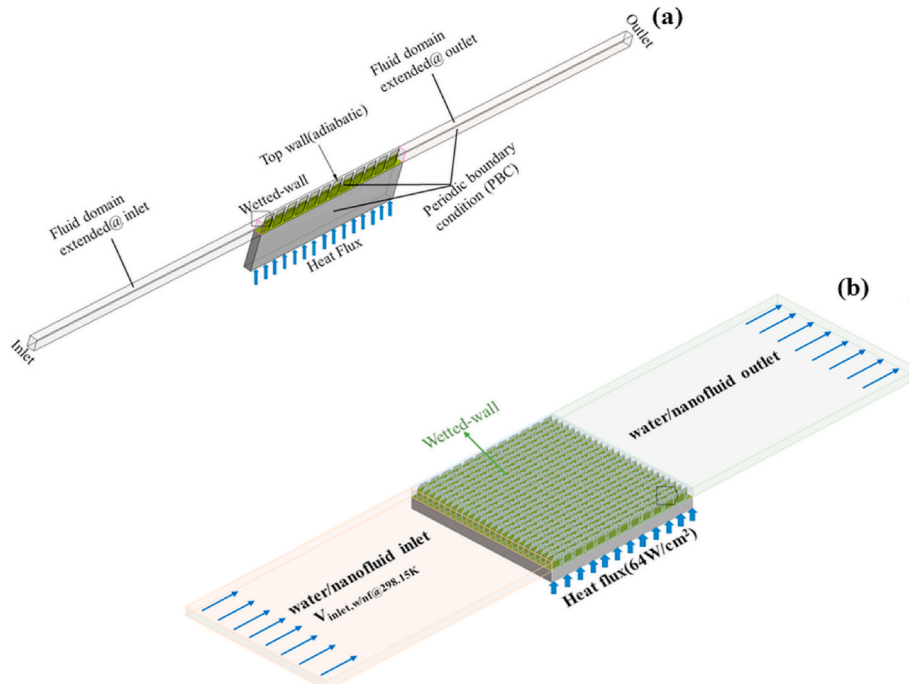


Fig. 2. (a) Periodic single domain model with boundaries (b) OFHS (Full domain) with boundary conditions: Isometric view.

3.4. Solution grid and mesh independence test

Grid independence is carried out for four separate grids to verify solutions, ensuring that exact results are obtained during the shortest measuring period, and the best measuring solution is found. The OFHS (periodic and full-domain model) computational mesh grid shown in Fig. 3 was created using ANSYS mesh.

To ensure $y^+ < 0.5$ and satisfy the enhanced wall treatment condition, the first layer cell thickness of 0.001 mm was considered with the mesh of 15 growing boundary layers. This is a prerequisite for adequately resolving the viscous boundary layer using the selected RNG k- ϵ model and enhanced wall treatment (EWT) [48]. Initially, the mesh is first defined with the following parameters on top surfaces of the fluid (water/nanofluid) domain: face sizing: 0.025 mm; fluid domain length sizing: 500 (no of division); fluid domain extension part sizing: 80 (no of division); fluid domain width sizing: 20 (no of division). Mapped face sizing is adopted for the entire domain. Furthermore, the mesh profile created for the fluid domain top surface was swept to the fluid domain bottom surface with 60 numbers of total division with a finer resolution near the top and bottom surfaces, serving as a boundary layer mesh to comply with y^+ criteria. The solid domain has meshed with the body sizing of 0.05 mm with the multizone-Hexa scheme. The top face of the fin has meshed with 0.04 mm sizing. A similar strategy as mentioned above is adopted for the meshing of the full domain model. The total number of mesh elements for the periodic domain is around 4.5 million (4,561,530), whereas, for the full domain model, it is about 22.5 million (22,318,077). Table 1 shows the mesh independence study of the present numerical investigation. The study has been conducted for OFHS on four grid sizes for $Re = 300$ at 0.5 volumetric concentration. Pressure drop and wetted wall temperature (T_{wetted}) are the evaluation

parameters for mesh independence study. We can see that the difference in the T_{wetted} is less than 0.1 K, and the variation in the pressure drop is less than 0.2 % compared to the most refined mesh. Therefore, 4.5 million mesh size criteria for the following design assessment are applied to consider the balance of numerical precision and computer time.

3.5. Setting implemented in solver

The governing equations and boundary conditions were numerically solved using commercially used ANSYS FLUENT 19.0. The double-precision solution scheme is adopted. The SIMPLEC algorithm was used for pressure velocity coupling, and a second-order upwind scheme was chosen to discretize the variables [51]. Except for energy, which is 10^{-10} , the convergence criteria of 10^{-6} are used. The periodic domain simulations were completed in 1–2 days for each case. In contrast, simulations of the full domain models were completed in 3–4 days for each case, using an Intel Xeon(R) E5-2620v4 processor with eight cores running at 2.10 GHz. At higher flow rates, the simulations converged more quickly.

4. Nanofluid preparation method and modeling

The experimental investigation uses Al_2O_3 nanoparticles suspended in water at four different volumetric concentrations of 0.5 %, 1 %, 1.5 % and 2 %. Nanoparticles used in this work were commercially purchased from Sigma-Aldrich (CAS number: 1344-28-1), and the characteristics of these nanoparticles are shown in Table 2 [53]. In a CFD simulation, these properties are used to calculate the thermal properties of an analogous single-phase fluid. The nanofluid is the technical term for this.

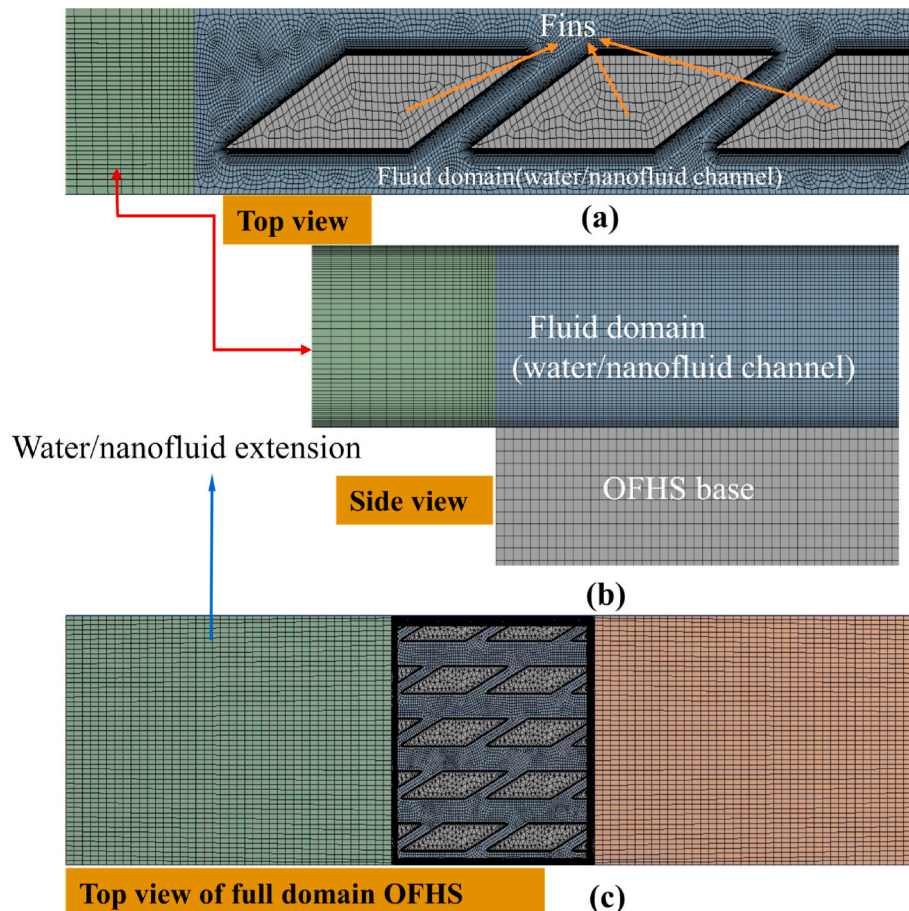


Fig. 3. Meshing of OFHS (a) Top View of periodic model (b) Side View periodic model (c) Top View of the full domain model.

Table 1

Mesh independence study.

Re	Heat flux (W/cm ²)	Pressure drop (Pa)	% difference	T _{wetted-wall} (K)	% difference	Total no of elements
300	64	734.18327	5.99 %	321.59967	0.038 %	2,174,970
300	64	706.3185	1.97 %	321.51372	0.011 %	3,256,110
300	64	693.38297	0.10 %	321.48313	0.002 %	4,561,530
300	64	692.6698		321.47662		6,242,430

Table 2Characteristic Al₂O₃ nanoparticle.

S. No.	Characteristics	Al ₂ O ₃
1	Size	≤13 nm
1	Colour	white
2	Purity	99.8 %
3	Density	3.95 g/cm ³
4	Sp. Heat	776.42 J/kgK
5	Thermal conductivity	37.717 W/mK
4	Sp. Surface area	85–115 m ² /g
5	Morphology (from TEM)	spherical

Table 3

Thermophysical properties of copper [44,48].

HS material	ρ (kg/m ³)	Cp (J/kgK)	k (W/mK)
Copper	8978	381	387.6

Table 3 lists the copper heat sinks' thermo-physical characteristics.

The process of nanofluid preparation is critical in measuring the thermal characteristics of nanofluid flows. The stability and homogeneity of nanoparticles in the base fluid are essential factors to avoid sedimentation and channel blockage. To avoid the negative impacts of clogging and sedimentation, nanofluids have been prepared with great care to maintain stability and homogeneity. To produce nanofluid, a two-step technique [54] was used in this study. The step-by-step process of the two-step technique has been depicted in Fig. 4. Initially, Al₂O₃ nanoparticles are precisely weighed using a precision balance (Sartorius, the accuracy of 0.01 mg). The weighted nanoparticles are then dissolved in the base fluid (DI water) to create nanofluid with the desired

volumetric concentrations. To homogenize the mixture, 1/2 h of magnetic stirring was applied using a SPINOT-Digital magnetic stirrer hot plate (Tarsons 6040). The nanofluid samples were placed in the water bath ultrasonicator after magnetic stirring. The fluid is ultra-sonicated for a minimum of 4–5 h using an ultrasonic cleaner (Model: Micro clean 109, Oscar) in 30 kHz of frequency. For a time period of 24–48 h, the produced nanofluids were found to be free of apparent sedimentation and clogging. Because the experimental run duration is smaller than the period specified for one set of flow rates, the nanofluids employed in the experiments may be said to be stable and evenly disseminated in the base fluid.

TEM images of nanofluids used in the experiments were also recorded to ensure that the nanoparticles are not visibly agglomerated and that the solution remains stable. TEM of Al₂O₃/water-based nanofluid has been shown in Fig. 5. The TEM image shows an almost uniform distribution of suspended nanoparticles in the base fluid. TEM results present the same average particle size as per the supplier information.

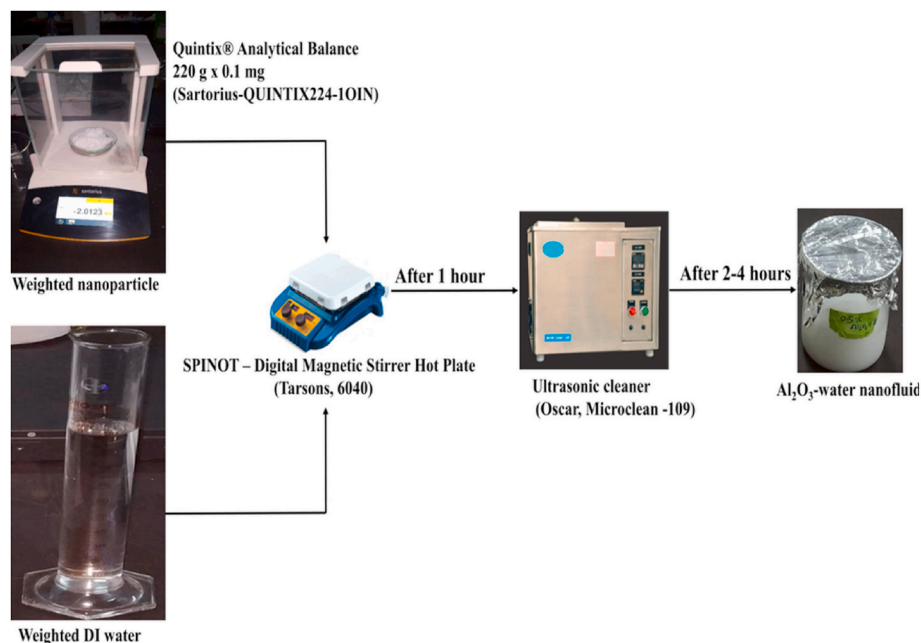
In the reported work, Al₂O₃/water nanofluids of different volume concentrations (0.5, 1, 1.5, and 2 %) have been employed. The thermophysical properties of nanofluids play a significant role in the simulation procedure since they strongly affect the computational results. The Hamilton Crosser model [55] provided the theoretical value of thermal conductivity, as given in (13)

$$k_{eff} = k_f \frac{[k_p + (n-1)k_f - (n-1)\varphi(k_f - k_p)]}{[k_p + (n-1)k_f + \varphi(k_f - k_p)]} \quad (13)$$

Eq. (13) is reduced to Eq. (14) for an alumina nanoparticle (spherical)

$$k_{eff} = k_f (1 + 3\varphi) \quad (14)$$

Thermal conductivity is derived from experimental findings

**Fig. 4.** Nanofluid preparation method (Two-step method).

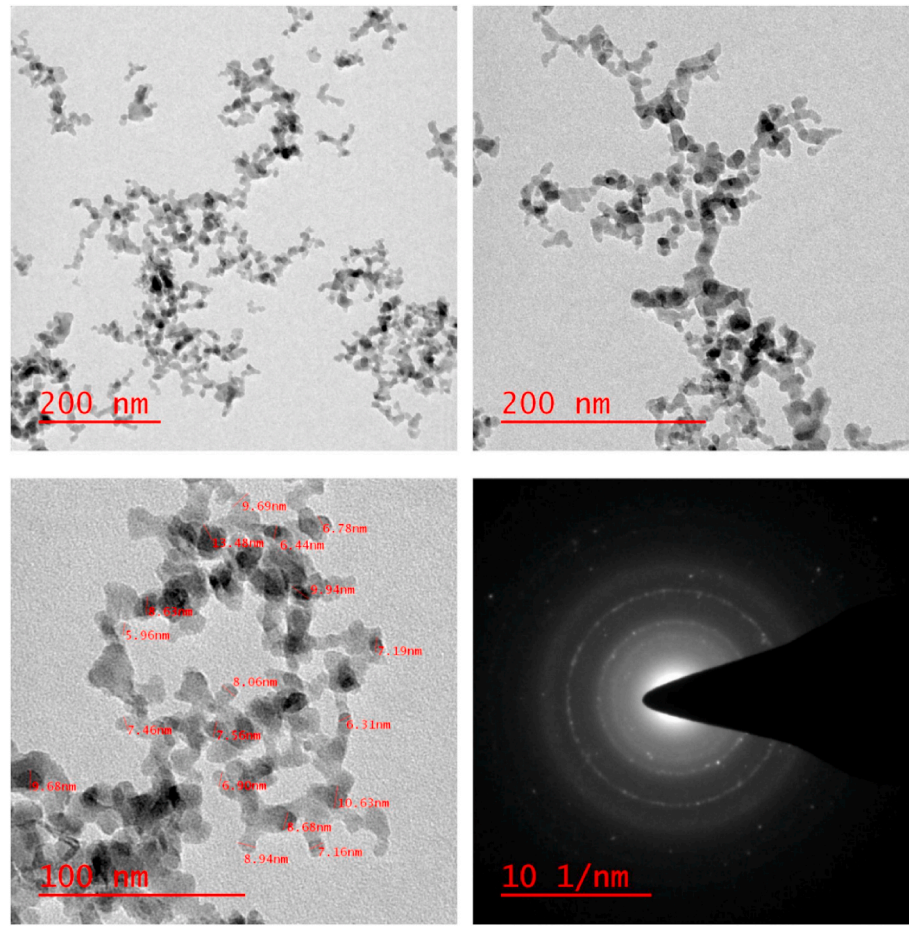


Fig. 5. TEM image of nanofluids samples.

available for Al_2O_3 -water nanofluid in the literature listed by Singh et al. [16,56].

$$k_{\text{eff}} = k_f(1 + 4\phi) \quad (15)$$

Also, by using the experimental data of Al_2O_3 -water nanofluid, Singh et al. [56] modified the Einstein model for viscosity as:

$$\mu_{\text{eff}} = \mu_f(1 + 10\phi) \quad (16)$$

The effective density and specific heat of nanofluids are calculated as in (17) and (18) [57]:

$$\rho_{\text{eff}} = \rho_p\phi + \rho_f(1 - \phi) \quad (17)$$

$$C_{p,\text{eff}} = \frac{[\rho_f C_{pf}(1 - \phi) + \rho_p C_{pp}\phi]}{\rho_p\phi + \rho_f(1 - \phi)} \quad (18)$$

The subscripts f, nf, and p stand for base fluid, nanofluid, and particle, respectively. The estimated value of nanofluid thermophysical properties is shown in Table 4.

Table 4
Thermo-physical properties of Al_2O_3 -water nanofluid.

Volumetric Conc.	Density (kg/m^3)	Heat capacity (J/kgK)	Thermal conductivity (W/mK)	Viscosity (Pa.s)
(0.5 %)	1012.04	4114.875	0.6191	0.0009428
(1.0 %)	1027.03	4051.643	0.6312	0.0009877
(1.5 %)	1042.02	3990.23	0.6434	0.0010326
(2.0 %)	1057.01	3930.559	0.6555	0.0010775

4.1. Procedure to obtain data in numerical analysis

The computational result data is post-processed to obtain parameters for the performance evaluation and quantitatively analyze thermo-fluidic characteristics. The area-weighted average is used to obtain the average wetted wall temperature ($T_{\text{wet, avg}}$). The water/nanofluid inlet and outlet temperatures ($T_{\text{w/nf inlet}}$, $T_{\text{w/nf outlet}}$) of the SCHS/OFHS were calculated using the mass-weighted average method. The pressure at the inlet and outlet of the HS ($P_{\text{w/nf inlet}}$, $P_{\text{w/nf outlet}}$) were also calculated based on mass-weighted. The term ξ (area and mass-weighted parameter) is calculated through the ANSYS FLUENT 19.0.

$$\xi_{\text{mass-weighted}} = \frac{\sum_{i=1}^{i=n} \xi_i \rho_i \left| \vec{A}_i \vec{V}_i \right|}{\sum_{i=1}^{i=n} \rho_i \left| \vec{A}_i \vec{V}_i \right|} \quad (19)$$

$$\xi_{\text{area-weighted}} = \frac{\sum_{i=1}^{i=n} \xi_i A_i}{A} \quad (20)$$

ξ is called target parameters, n denotes no of cells, and V_i is the velocity vector at the respective boundary.

The average Reynolds number and hydraulic diameter (D_h) are calculated for both geometries as in Eq. (21) and Eq. (23).

$$\text{Re} = \frac{\rho V_{\text{avg}} D_h}{\mu_{\text{at } T_{\text{m,w/nf}}}} \quad (21)$$

The mean water/nanofluid temperature is given as in (22)

$$T_{m,w/nf} = \frac{T_{w/nf,inlet} + T_{w/nf,outlet}}{2} \quad (22)$$

$$D_h = \frac{4W_{ch}H_{ch}}{2(W_{ch} + H_{ch})} \quad (23)$$

V_{avg} is the average water/nanofluid velocity inside the channels. Due to the presence of fin inside the single domain model for SCHS/OFHS, the velocity is different at the SCHS/OFHS inlet (V_{inlet}) from the velocity defined at the inlet boundary. The velocity at the inlet is defined as:

$$V_{avg} = \frac{W_{SD}}{W_{ch}} V_{in} = \frac{1}{0.5} = 2V_{in} \quad (24)$$

For the single domain of channels, the volumetric flow rate is obtained from the following formula

$$\dot{V}_{o/w/nf} = \sum_{i=1}^n A_i V_i \text{ (For single domain simulation)} \quad (25)$$

$$\dot{V}_{o/w/nf} = \sum_{i=1}^n A_i V_i \times \dots \text{no of channels (For full domain and experiments)} \quad (26)$$

The mass flow rate is calculated as in Eq. (17).

$$\dot{m}_{w/nf@inlet} = \sum_{i=1}^n \rho_i A_i V_i \text{ (For single domain OFHS)} \quad (27)$$

$$\dot{m}_{w/nf@inlet} = \sum_{i=1}^n \rho_i A_i V_i \times \dots \text{no of channels (For full domain and Experiments)} \quad (28)$$

Pressure drop over the entire domain is measured as in (29) (including inlet and outlet extension)

$$\Delta P = P_{w/nf,inlet} - P_{w/nf,outlet} \quad (29)$$

Eq. (30) defines the average heat transfer coefficient

$$h_{avg} = \frac{q_{hb}}{A_{wetted} [T_{wetted,avg} - T_{w/nf,m}]} \quad (30)$$

where q_{hb} power supplied in the simulations.

The average Nusselt number (Nu_{avg}) and Fanning Friction Factor (f_F) are given in Eq. (31) and Eq. (32).

$$Nu_{avg} = \frac{h_{avg} D_h}{k_{@T_{m,w/nf}}} \quad (31)$$

$$f_F = \frac{\Delta P D_h}{2\rho_{@T_{m,w/nf}} V_{avg}^2 L_{hs}} \quad (32)$$

5. Experimental investigation

The thermofluidic performances of the Straight channel heat sink (SCHS) and oblique fin heat sink (OFHS) were measured using the given experimental flow loop. OFHS and SCHS were fabricated by CNC machining and tested experimentally on the single-phase heat transfer loop to validate the simulation findings. The exploratory analysis will be discussed in depth in this section.

5.1. Experimental setup (flow loop arrangement)

The schematic diagram of the water/nanofluid flow loop is shown in Fig. 6. The flow loop consists of several components: liquid-to-air heat exchanger, valves, reservoir tank, gear pump, filter, flow meter, differential pressure transmitter, DC power supply, data acquisition system, computer, thermocouples, and the test section. Fig. 7 shows the actual fabricated experimental water/nanofluid loop. The loop contains the entire component, as discussed earlier. The utmost care has been taken to fabricate the loop in our laboratory.

DI-water/nanofluid is stored in the reservoir tank. In the test operation, with the help of the flow loop, the fluid is sent to the test section via a micro gear pump through the 15- μ m filter. The flow meter has a measurement range of 100–1000 ml/min to measure the flow rate of the fluid. A bypass line is provided at the exit of a gear pump to control the

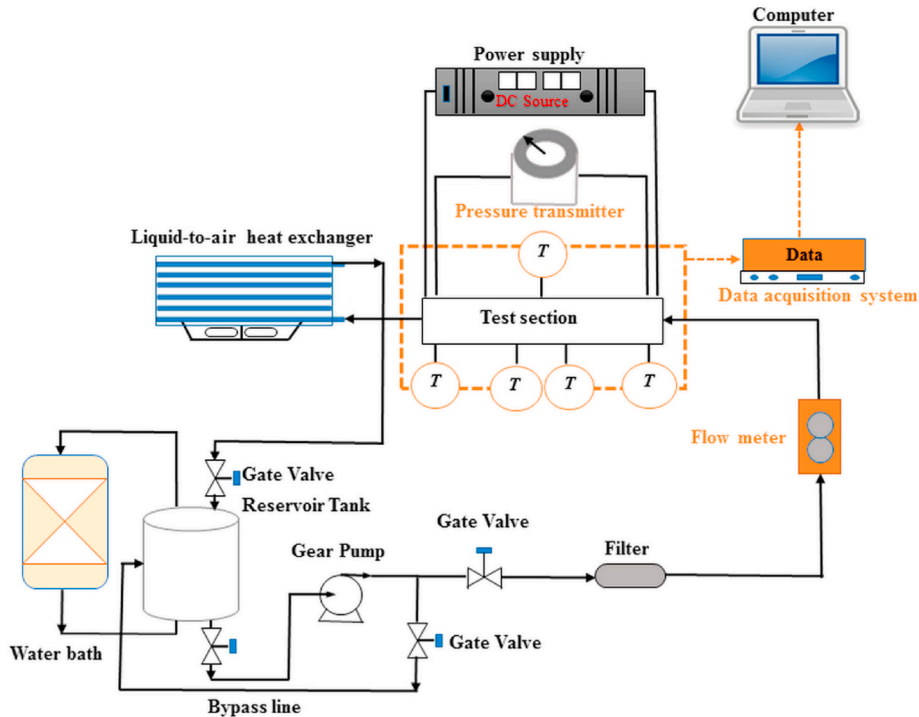


Fig. 6. Schematic view of the experimental setup.

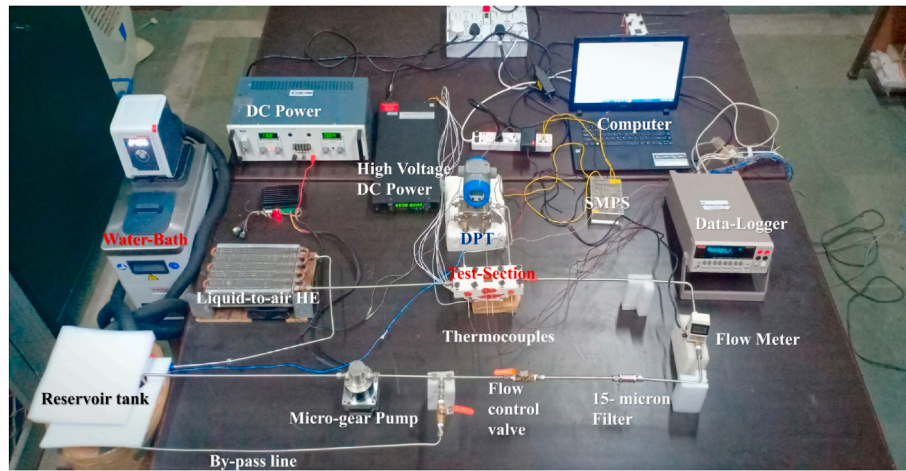


Fig. 7. Photograph of the experimental arrangement.

desired flow rate in the test section, and the remaining portion is sent back to the reservoir tank. The temperature measurement is carried out with a T-type probe thermocouple at the inlet and outlet plenum of the test section and another three locations (below 3.25 mm of the channel wall) of the microchannel heat sink. The pressure drop between the inlet and outlet plenum is directly measured by using a differential pressure transmitter. The heating of the test section is provided through the four-cartridge heater, which is powered by a high voltage DC power source. A liquid-to-air heat exchanger is provided to cool the DI water/nanofluid after passed through the test section before entering into the reservoir tank. All the pipe and fitting used to construct the flow loop is made up of Swagelok to ensure the leak-proof fitting for the smoothness of the experiment. A high-speed data acquisition system collects the data from all thermocouple sensors. All the equipment used in the investigation is listed in Table 5.

5.2. Microchannel test-section with the setup used for investigation

The most significant aspect of the experimental work is the design of the test section (heat sink with its housing). The SCHS (Fig. 8) and OFHS (Fig. 9) HSs are fabricated on the top of the copper block of a 25 × 25 mm footprint.

At 26.5 mm below the bottom surface of the microchannel, an extended part (10 mm on each side) known as the flange is fabricated to fix the position of the copper block in the direction of height when inserted into the top housing. For the provision of cartridge heaters, four holes of 8 mm diameter and 38.5 mm length from the bottom portion are

created, shown in Figs. 8 and 9.

A 3D optical profilometer is used to measure the surface roughness of the OFHS. Fig. 10 depicts the surface roughness as well as the 3D profile. Roughness parameters such as fin length and fin pitch are measured and presented inside the table of Fig. 10. The roughness parameter values have been introduced. The measured average surface roughness (Ra) is 4.76 μm.

The Test section housing comprises four parts: the base, the bottom housing, the top housing, and the top cover (Fig. 11 (a)). All the housing is made of Teflon, whereas the top cover comprises acrylic sheet material to visualize the flow inside the test section. The microchannel heat sink rests in the top housing. The top housing consists deep plenum at the inlet and outlet, and shallow plenums are present after and before the copper block, which is 40 mm long to satisfy the extension given at computational models at the inlet and outlet. With the provision of four holes; where two holes are used for the T-type probe thermocouple, and another two holes are used to provide the differential pressure transmitter. In the heat sink, three holes were drilled at 3.25 mm below the surface for thermocouples to measure the temperature of the surface along a streamwise direction. Four holes were also drilled at the bottom of the copper block for the provision of cartridge heaters to provide the heat flux. Finally, the top cover is tightly screwed with Allen key screws, and the entire assembly is sealed with the help of ANABOND high-temperature sealant to prevent any leakage. On the other side, the whole assembly rests on the base for the smooth flow of fluid and stability of the system. Fig. 11 (b) depicts the assembled view of the test section, which is ready to fit inside the experimental loop.

5.3. Procedure adopted in experiment

5.3.1. Heat loss calculation

Once the test section is installed inside the flow loop, the calculation of heat loss is necessary. The proper fitting and sealing of the test section cannot prevent heat loss. Therefore, before starting the experiment, heat loss is calculated immediately after the test part assembles in the flow loop [38]. Since the ambient temperature is recorded simultaneously, the heat loss is obtained by measuring the temperature of heat sinks at variable heater power without the fluid flow. The heat loss formulation was presented in the form of a linear equation concerning the temperature difference between heat sink temperatures and ambient (environment). Fig. 12 depicts the heat loss best-fit curve. The heat loss calculated with the help of correlation in the linear form can be given in Eq. (33).

The function defined for the analysis of heat loss is:

$$Q_{loss} = 0.11462(T_{HS,avg} - T_{amb}) - 0.22576 \quad (33)$$

Table 5

List of equipment's and sensor used in the experiment.

Si No	Equipment	Model	Make	Specifications
1	Reservoir tank	–	Aimil Ltd	10 Liter
2	Micro gear pump	MG312XK/24W	Topsflow	500–5000 ml/min
3	Filter	SS-4F-7	Swagelok	15 μm
4	Digital Flow meter	FLR1010D	Omega	100–1000 ml/min
5	Differential pressure Transmitter	STD-720	Honeywell	–
6	Thermocouples	T-type	Watlow	–
7	RTD	PT100	Watlow	–
8	DC Power Supply	XG 300–5.6	Ametek	0–150 V, 0–5.6A
9	Data Acquisition System	2701	Keithley	–
10	Liquid-to-air heat exchanger	–	Aimil Ltd	–
11	Water Bath	CD200F	Julabo	–

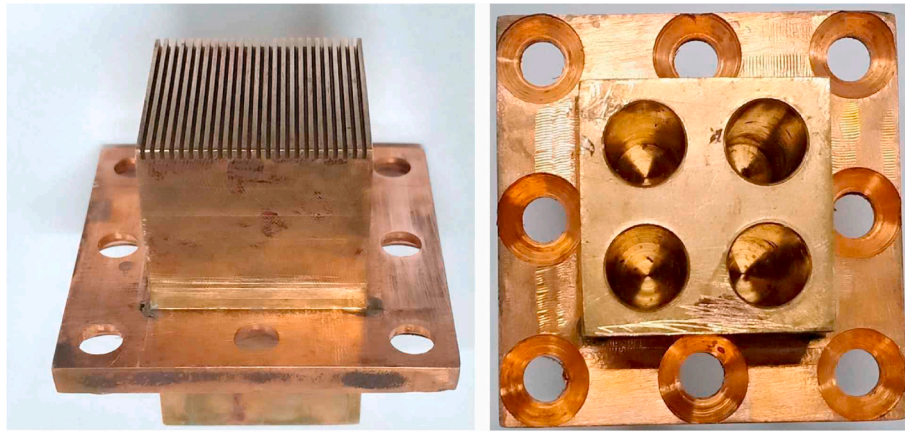


Fig. 8. Fabricated SCHS sink used in the experiment.

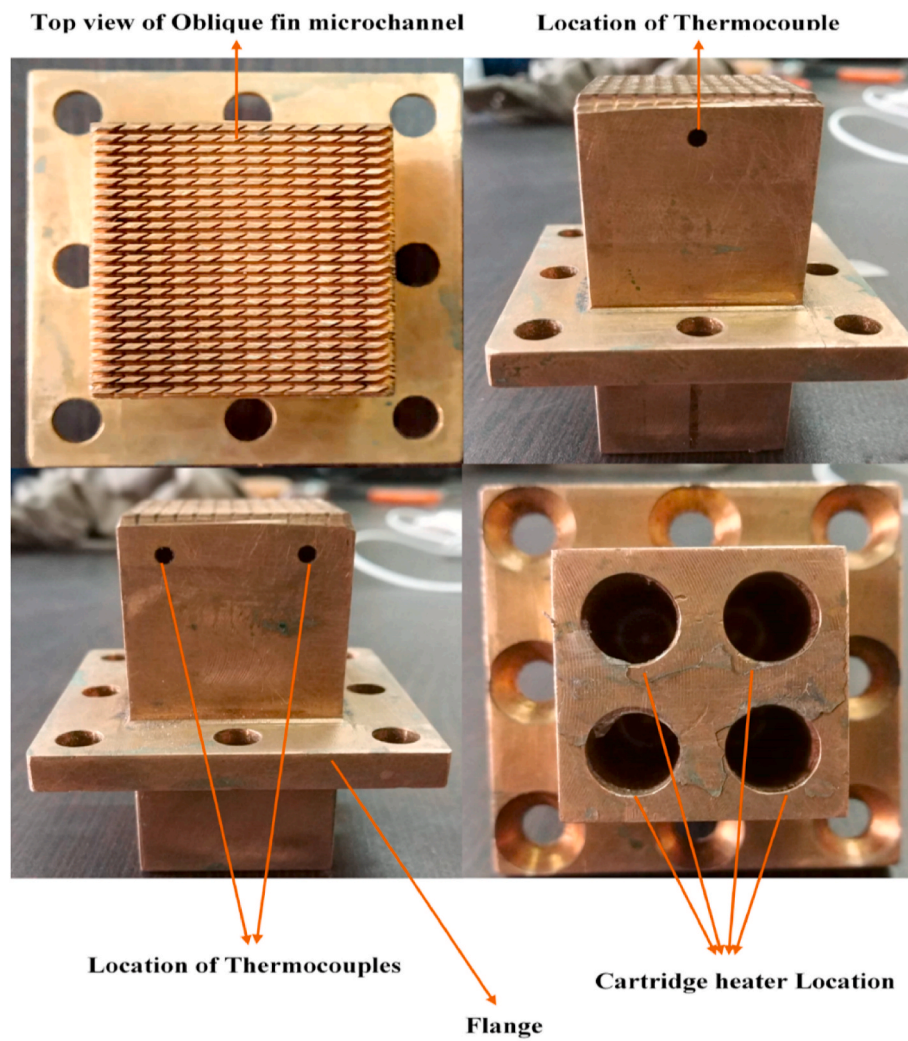


Fig. 9. Fabricated Oblique fin heat sink used in the experiment with the description of a different location.

The effective heat dissipated through the water/nanofluid is obtained through Eq. (34)

$$Q_{w/nf, effective} = V_{heater} I_{heater} - Q_{loss} \quad (34)$$

5.3.2. An operation carried out in the flow loop

Once the heat loss calculation is done, the following operations have

been performed in the flow loop during the experiments:

1. All the gate valves are in open condition before starting the gear pump.
2. The required flow rate is achieved by adjusting the speed of the pump and valve in the bypass line.

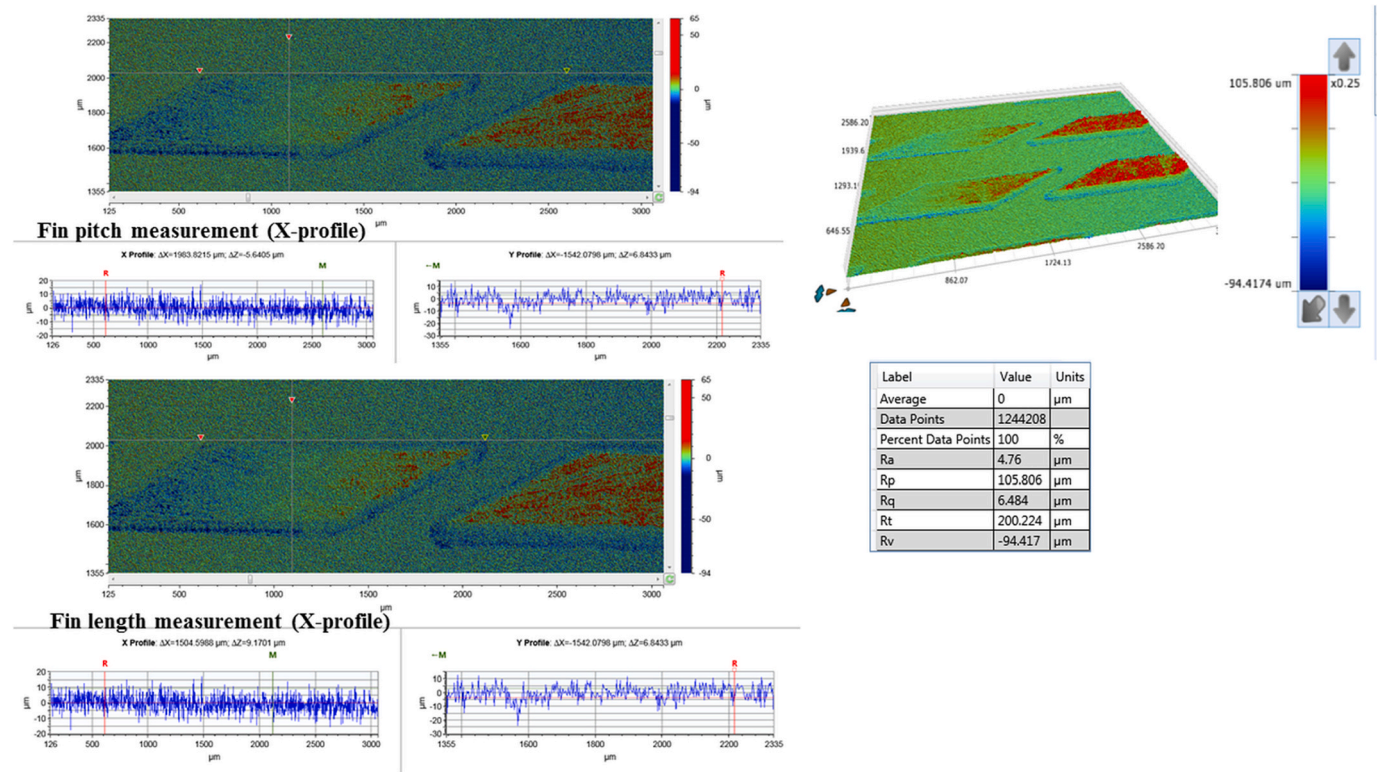


Fig. 10. 3D surface measurement of OFHS MC.

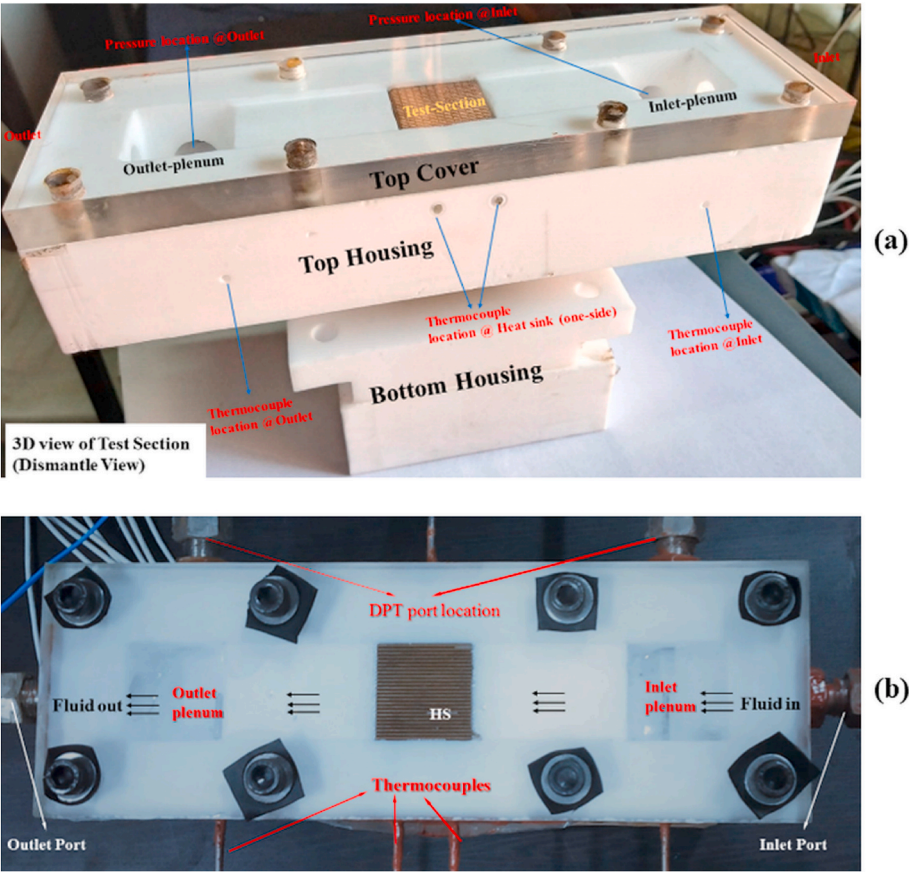


Fig. 11. (a) Dismantle view of Fabricated Test-section with housing used in the experiment with the description of the different location (b) Assembled test section.

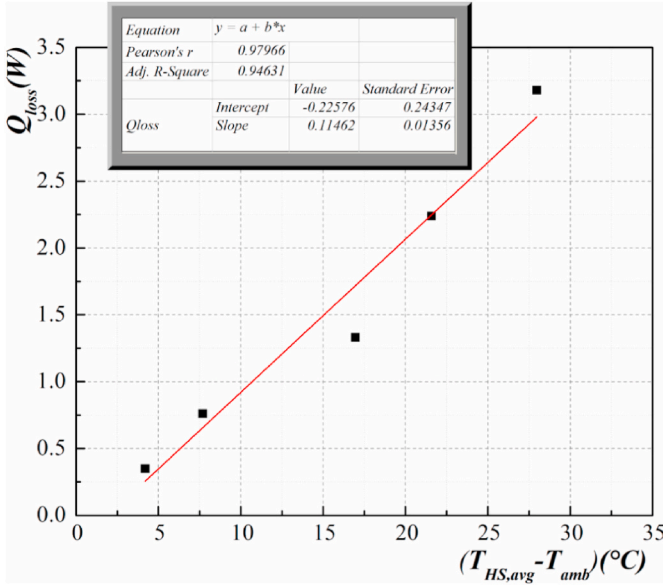


Fig. 12. Heat loss calculation with best-fit curve.

- Once the flow rate is stable, the desired power is supplied to the cartridge heaters through the high voltage DC power supply to get the desired $Q_{water/nf}$.
- The water bath is placed to adjust the inlet temperature of water and nanofluid.
- Finally, once the stable condition is reached, all the data is recorded through the data logger for 1 min.
- The flow rate is then changed for the next flow rate, and the entire process 1 to 5 repeats.

5.3.3. Error estimation and analysis

A systemic uncertainty analysis is necessary to estimate errors related to the experiments in micro and mini-experimental studies. Taylor [60] performs standard error analysis.

$$\delta R = \sqrt{\sum_{j=1}^M \left(\frac{\partial R}{\partial X_j} \delta X_j \right)^2} \quad (35)$$

where δX_j shows the uncertainty in the j th measured parameter. The calculated value of uncertainties has been depicted in Table 6.

Manufacturers/suppliers are responsible for the uncertainty of sensors and instruments. The sensor uncertainties are set out in Table 7.

5.3.4. Experimental data analysis

Once the experiment has started, the following data has been recorded through a data logger:

- The average temperature of the heat sink ($T_{hs, avg}$),
- the average temperature of the fluid (water/nanofluid) at the heat sink inlet and outlet ($T_{w/nf, inlet}$, $T_{w/nf, outlet}$)

The term (1) and (2) has been calculated using the equation below.

Table 6
Uncertainties range.

Defined parameter	Uncertainty range
Pressure drop	0.04995 %
Re	3.7783 %
h_{avg}	4.9138 %
Nu_{avg}	5.7669 %
f_f	5.8486 %

Table 7

Uncertainties for instruments/sensors.

Type of Instruments/Sensors	Range of Uncertainty
Differential pressure transmitter (Honeywell)	0.05 %
Voltage and Current DC Power Supply (AMTEK-Sorenson)	$\pm 0.05\%$
Load regulation (constant voltage mode)	
T-Type thermocouples (Probe)	$\pm 0.5^\circ \text{C}$
Flow Meter/Sensor (Omega)	$\pm 1\%$ of Full Scale for liquid

$$T_i = \frac{1}{n} \left(\sum_{j=1}^n T_{i(j)} \right) \quad (36)$$

where n is the number of thermocouples used for each reading. The heating power given to the cartridge heaters used in the experiment is directly recorded from the high voltage DC power supply.

In the experiments, the temp of the heat sink was 3.25 mm below the fin base (microchannel bottom surface). The wetted wall temperature (T_{wetted}) is calculated as in (37);

$$T_{wetted-wall} = T_{hs} - \frac{d_{hs} Q_{w/nf, effective}}{k_{hs} A_{hs, base}} \quad (37)$$

d_{hs} = distance between from heat sink base (fin base) to thermocouple location.

Therefore, the junction temperature (T_j) is calculated at 12 mm below the fin/channel base using the 1-D heat conduction assumption.

$$T_{j, avg} = T_{hs, avg} + \frac{H_{hs} Q_{w/nf, effective}}{2k_{hs} A_{hs, base}} \quad (38)$$

The mass flow rate of the water/nanofluid is then calculated as

$$\dot{m}_{w/nf} = \rho_{@T_{w/nf, inlet}} \dot{V}_{w/nf} \quad (39)$$

The term $\dot{V}_{w/nf}$ is the volumetric flow rate measured directly from the flow meter.

The average heat transfer coefficient is defined as:

$$h_{avg} = \frac{Q_{w/nf, effective}}{A_{Wetted-wall} (T_{wetted} - T_{w/nf, mean})} \quad (40)$$

Ultimately, the Nusselt number and fanning friction factor are calculated as in (31) and (32).

5.3.5. Explanation of pressure drop calculation

The pressure difference recorded by the differential pressure transducer (DPT) between inlet plenum to outlet plenum is shown in Fig. 13, which can be the combination of.

- (1) Contraction loss from inlet plenum to the shallow plenum, ΔP_{c1}
- (2) Frictional loss in the shallow plenum at the inlet, $\Delta P_{friction,1}$
- (3) Contraction loss from inlet shallow plenum to heat sink inlet, ΔP_{c2}
- (4) Frictional loss inside the SCHS and OFHS MC, $\Delta P_{friction,2}$
- (5) Expansion from the heat sink to the shallow plenum at the outlet of MC, $\Delta P_{expansion,2}$
- (6) Frictional loss in the outlet shallow plenum, $\Delta P_{friction,3}$
- (7) Expansion from shallow to outlet plenum, $\Delta P_{expansion,1}$.

$$\Delta P_{Recorded(DPT)} = \Delta P_{contraction,1} + \Delta P_{friction,1} + \Delta P_{contraction,2} + \Delta P_{friction,2} + \Delta P_{expansion,2} + \Delta P_{friction,3} + \Delta P_{expansion,1} \quad (41)$$

In the simulation, the inlet and outlet are extended to match the experimental conditions. Therefore, all the terms are neglected except the first and last term. Thus, the pressure drop across the SCHS/OFHS MC is calculated as:

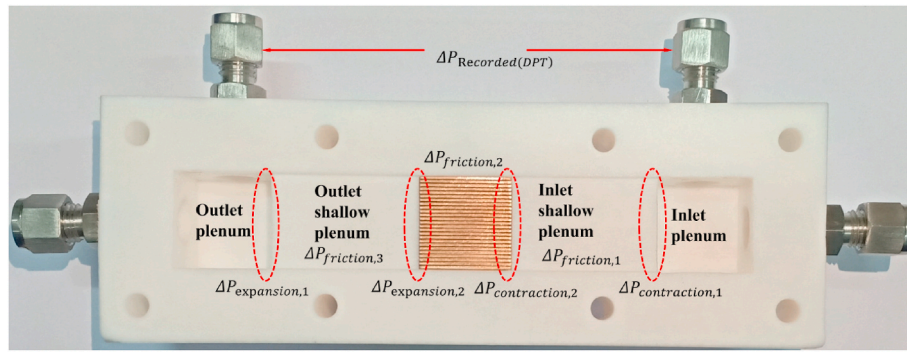


Fig. 13. Description of a different location for calculation of experimental pressure drop.

$$\Delta P_{SCHS/OFHS} = \Delta P_{Recorded(DPT)} - (\Delta P_{contraction,1} + \Delta P_{expansion,1}) \quad (42)$$

The contraction and expansion losses can be estimated as given in Ref. [58].

$$\Delta P_{contraction,1} = \frac{1}{2} \rho_{w/nf} (u_{SP,inlet}^2 - u_{DP,inlet}^2) + \frac{1}{2} K_{contraction,1} \rho_{w/nf} u_{SP,inlet}^2 \quad (43)$$

$$\Delta P_{contraction,2} = \frac{1}{2} \rho_{w/nf} (u_{DP,outlet}^2 - u_{SP,inlet}^2) + \frac{1}{2} K_{expansion,1} \rho_{w/nf} u_{SP,outlet}^2 \quad (44)$$

The value of $K_{contraction,1}$ is assumed to be 1, and $K_{expansion}$ is calculated with the help of the following formula, as given in Eq. (45).

$$K_{expansion,1} = \left(1 - \frac{A_{SP}^2}{A_{DP}^2} \right)^2 \quad (45)$$

A_{SP} and A_{DP} is the cross-sectional area of the shallow and deep plenum, respectively. The velocities can be obtained by the ratio of the flow rate of water/nanofluid taken from the flow meter to the corresponding area.

6. Results and discussion (performance investigation and comparison)

6.1. Repeatability investigation and numerical validation

The numerical results were correlated with the experimental information to ensure the exactness of the numerical method used in this analysis. The experiment is intended to validate the results of 3D numerical analysis. The OFHS and SCHS were designed and tested experimentally in the fabricated experimental loop. Heat flux is kept constant

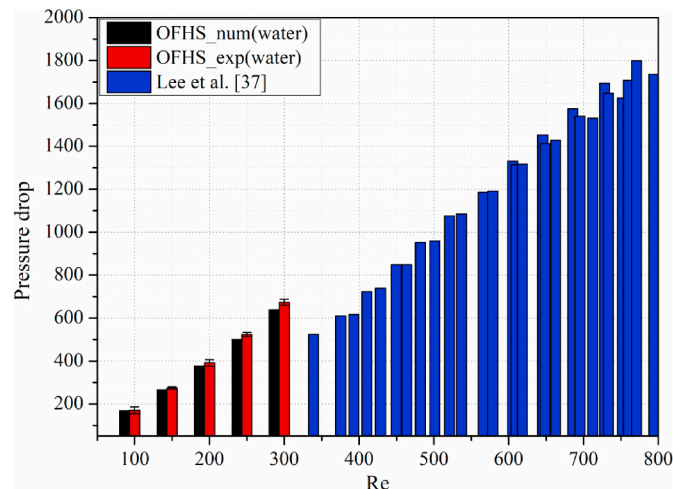


Fig. 14. Validation of experimental and 3D numerical results in pressure drop.

at 64 W/cm² as in simulations, although flow rates differ across the studied range. Fig. 14 shows a comparison of 3D simulation and experimental effects. The results for pressure drop agree reasonably well with the experimental data of Lee et al. [37]. Since these errors are deemed appropriate, the 3D simulation results are accepted as reliable data for further study. Finally, the complete set of tests was replicated on a different day to ensure that the test results were free of random errors. Fig. 15 shows the performance parameters from the two data sets for both heat sinks. The variations between each data sets were found in acceptable limits with experimental uncertainty.

6.2. The flow characteristic of nanofluid in oblique channel

The flow and temperature fields have been extensively studied to consider the thermal and hydraulic properties of Al₂O₃-water nanofluid flow through the OFHS. Because of the superior grid, periodic computational models with enhanced resolution of the single-channel were tested. Fig. 16, Fig. 17, and Fig. 18 display the pressure, temperature and velocity contours of OFHS, respectively. Since the results of the full domain simulation models of lower grid quality were already validated with the experimental results, the periodic simulation models of superior grid quality are supposed to provide accurate details regarding the flow and temperature fields of the OFHS. The contour plots were drawn at the channel mid-height at a 2 % concentration of Al₂O₃-water nanofluid, corresponding to Re = 300.

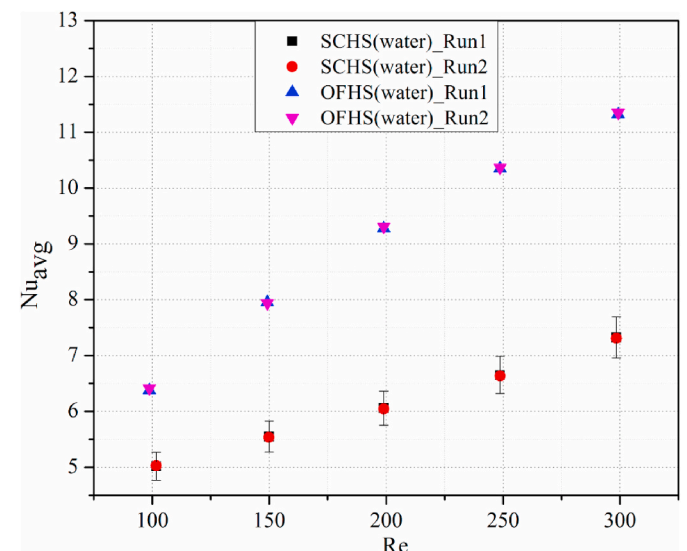


Fig. 15. Experimental repeatability investigation.

6.2.1. Pressure field analysis

One of the most desirable traits for OFHS is a slightly higher or negligible pressure drop penalty than SCHS. The pressure drop penalty depends on the secondary flow rates. The retention of flow momentum in another direction than the natural flow direction increases the pressure drop and enhances heat transfer. Fig. 16 (a) shows the pressure contour of OFHS in a streamwise direction. The formation of adverse pressure gradient has been noticed at the vicinity of the secondary location of the oblique fins, where the flow separation and boundary layer reinitialization occurs. As the Reynolds number is less, the effect of this adverse gradient is not significant. As Reynolds number is increased further, this effect contributes to the slightly higher pressure penalty but cannot underestimate heat transfer enhancement.

Fig. 16 (b) depicts the experimental pressure drop data through SCHS and OFHS. The simulation results agree very well with the experimental results. It is worth noting that the pressure drop in the OFHS for all Reynolds investigated is comparable to SCHS. This means that better heat transfer can be carried out without a significant pressure drop penalty. The opposing effects of increased pressure drops (because of the thinning and separation of the boundary layer) and pressure recovery make this possible (due to flow convergence and divergence). In comparison with the improvement in heat transfer, the pressure drop penalty is, therefore, minimal. The pressure drop increases with nanofluid concentration, but the same gear pump can manage it without occurring more pumping power (as pressure is less than 1 kPa).

6.2.2. Temperature field analysis

Fig. 17 reveals that the OFHS can retain an even/uniform fin temperature. The increase in heat transfer depends strongly on the secondary channel flow rates. This can be verified from the observation noted from the temperature and velocity field of OFHS. The temperature of the nanofluid within the OFHS secondary channels is found in a significant amount (as shown in the flow mixing plane). As a result, the advection part grows. By causing more extreme mixing flows due to the higher secondary nanofluid flow rate, the OFHS enables a more uniform accumulation of the fluid temperature in the streamwise direction.

6.2.3. Velocity field analysis

Fig. 18 depicts the velocity contour inside the OFHS. The flow rate in the primary channel of the OFHS is found higher than that of the

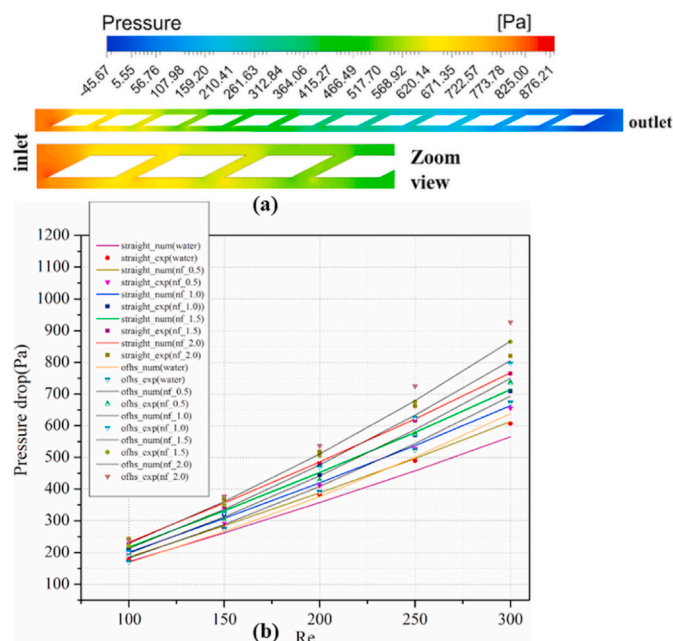


Fig. 16. (a) Pressure contour in the OFHS (b) Pressure drop Vs. Re

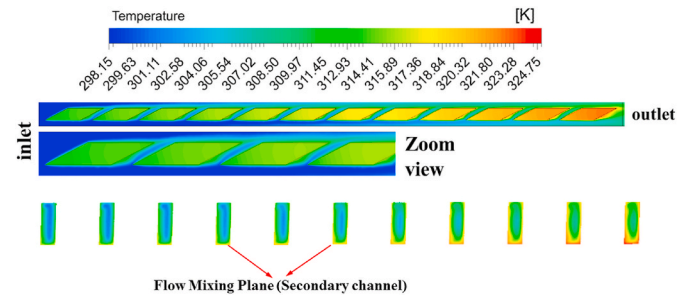


Fig. 17. Temperature contour.

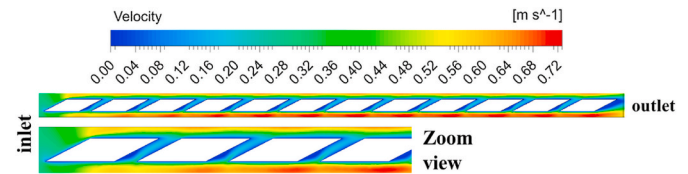


Fig. 18. Velocity contour.

secondary channel. The zoom view of Fig. 19 demonstrates a lower velocity flow field occupies a small section of the fin surfaces along the secondary flow portion of the channels. These lower-speed regions are the vortices caused by flow separation, discussed in the segment after this one. In the secondary channels of OFHS, the vortices are more diminutive, enabling higher secondary flow rates. Simulation findings show a clear difference in the area of flow of the OFHS. It is shown that the oblique fin disturbs the velocity profile at every entry of the downstream fin and, causes reinitialization of the hydrodynamic boundary layer at every oblique downstream fin. In contrast with the SCHS, the thickness of the boundary layer is considerably reduced. As a result, the velocity profile for the oblique fin structure is preserved in the developing region.

Figs. 19 and 20 show the velocity vector and streamline of the flow field. As fluid is forced into the OFHS, water/nanofluid goes inside the primary and secondary channels. At $Re = 100$ (low Reynolds number), the velocity vector and streamline are found in order. The low velocity inside the secondary channel (near the separation edge) and thus less fluid mixing is the reason behind this effect. Whereas at $Re = 300$, the flowing fluid over the separation edge causes the formation of vortices inside the secondary channel of the OFHS. This generates an adverse pressure gradient, which pulls the nanofluid flow back into the oblique channel from downstream of the fin. The high-temperature nanofluid is drawn back into the secondary channel, where it crosses the vortex in a circular way prior to re-joining the mainstream and extracted the high temperature from the fin wall.

The OFHS induces a vortex, as shown in Fig. 20, within the secondary

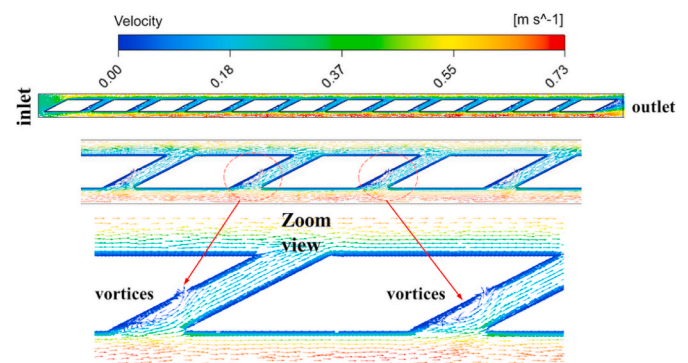


Fig. 19. Velocity vector inside the primary and secondary channel.

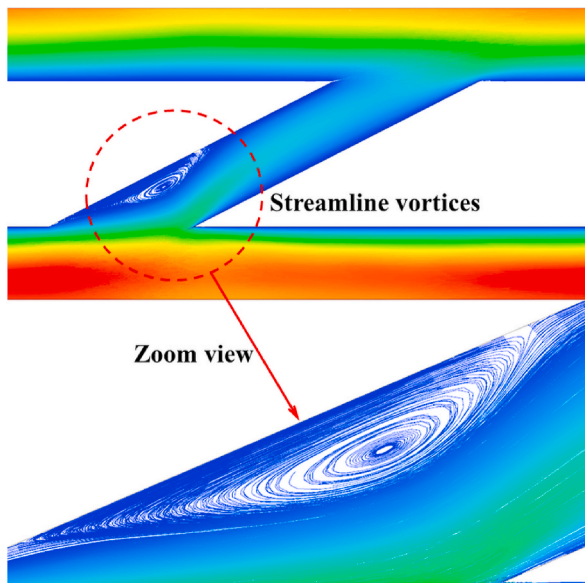


Fig. 20. Streamline visualization in the secondary channel.

channel only. Once the vortex is generated, It improves advection heat transfer and contributes to heat transfer performance, which shows slightly chaotic behaviour similar to phenomena reported by Dejong and Jacobi [59]. Hence, the heat transfer enhancement of the OFHS becomes more significant than its pressure drop penalty.

6.3. Heat transfer and friction factor characteristic

The experiment on heat sinks is carried out at the Reynolds numbers of 100–300. Fig. 21 depicts the average convection coefficient to Reynolds number for both SCHS and OFHS. The average convection coefficient for both designs generally increases with the Reynolds number as the thermal boundary layer thickness decreases with the increased fluid velocity. However, the heat transfer for the OFHS is found substantially higher compared with SCHS. The average convection coefficient increases by about 30 % at the lowest Reynolds number ($Re = 100$) and, by 55 % at a higher Reynolds number ($Re = 300$) for the volumetric concentration of 2 %. The combined effects of thermal boundary layer redevelopment at the edge of the oblique fin and the secondary flows produced by the flow diversion on the oblique channels are the reason for the appreciable increase in heat transfer. In addition, the inclusion of nanofluid also enhances the heat transfer rate because of its improved thermal conductivity. The heat transfer increase with the particle

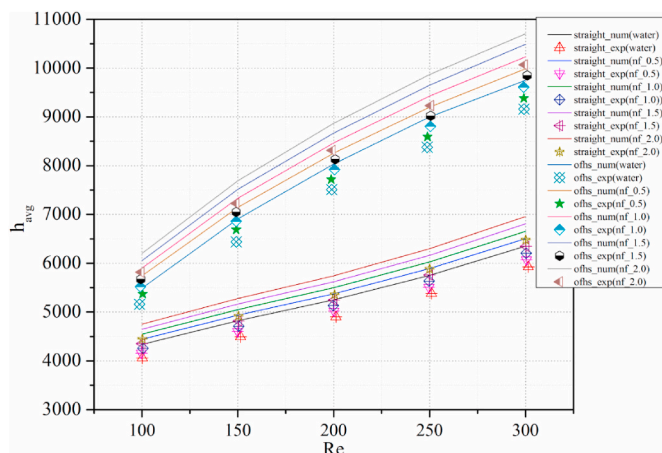


Fig. 21. Average convection coefficient vs. Reynolds number.

volume concentration. Increasing the effective heat transfer area over the SCHS produces a considerably lower temperature increase above the fluid inlet temperature for OFHS, combining highly increased convection coefficient.

A penalty in pumping powers always accompanies the enhancement in heat transfer. To illustrate the overall performance considering both heat transfer and pressure drop, a plot of friction factor versus Re is drawn, as shown in Fig. 22. It is evident that the average convection coefficient of OFHS is significant compared to SCHS. This enhancement effect is further obtained with the increment of Reynolds number and the volumetric concentration. However, the friction factors are almost equal in both designs due to the equal or slightly higher pressure penalty in OFHS than SCHS. In addition, the difference is nearly equal throughout the entire flow regime.

6.4. Thermal hydraulic performance factor

To analyze the thermal and hydraulic performance factors, the evaluation of two terms is important. (a) The heat transfer enhancement factor (E_{Nu}) and (b) pressure drop penalty (E_f). Both terms (a) and (b) are defined as the ratio of Nusselt number/and friction factor of OFHS to its corresponding SCHS and calculated numerically and experimentally for both HSS. From Fig. 23, the benefit of the OFHS becomes apparent where an increase in heat transfer enhancement factor exceeds the pressure drop penalty in all situations. The pressure drop penalty is equivalent to 1, which means the additional pressure drop is negligible or slightly more. Simultaneously, approximately 1.26–1.55 times enhancement in heat transfer is found in OFHS (Fig. 23). The heat transfer rate of the OFHS was about 50 % higher than the SCHS. The heat transfer enhancement factor can surpass as the Reynolds number increases. However, in the present experimental study, pressure drop penalty was observed less than 1 kPa, which is small and with the same gear pump is to be managed.

The factor E_{Nu} and E_f were calculated in Eq. (46) and Eq. (47), respectively.

$$E_{Nu}(Num / Exp) = \frac{Nu_{avg, OFHS}}{Nu_{avg, SCHS}} \bigg|_{Re} \quad (46)$$

$$E_f(Num / Exp) = \frac{f_{F, OFHS}}{f_{F, SCHS}} \bigg|_{Re} \quad (47)$$

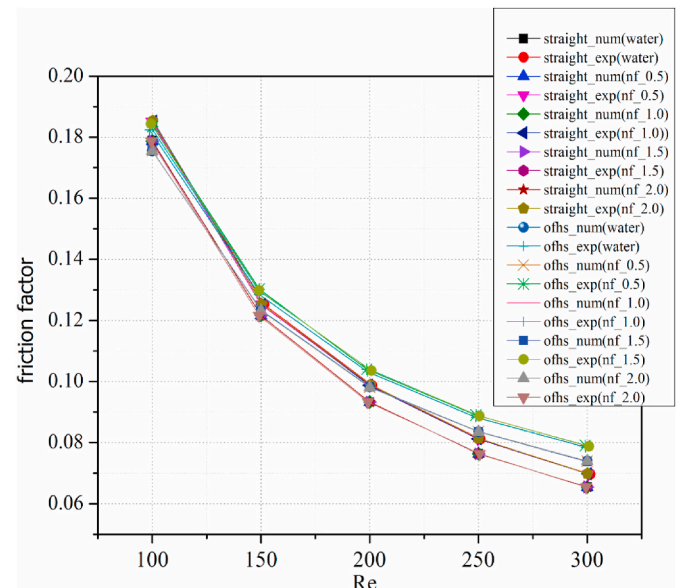


Fig. 22. Friction factor vs. Reynolds number.

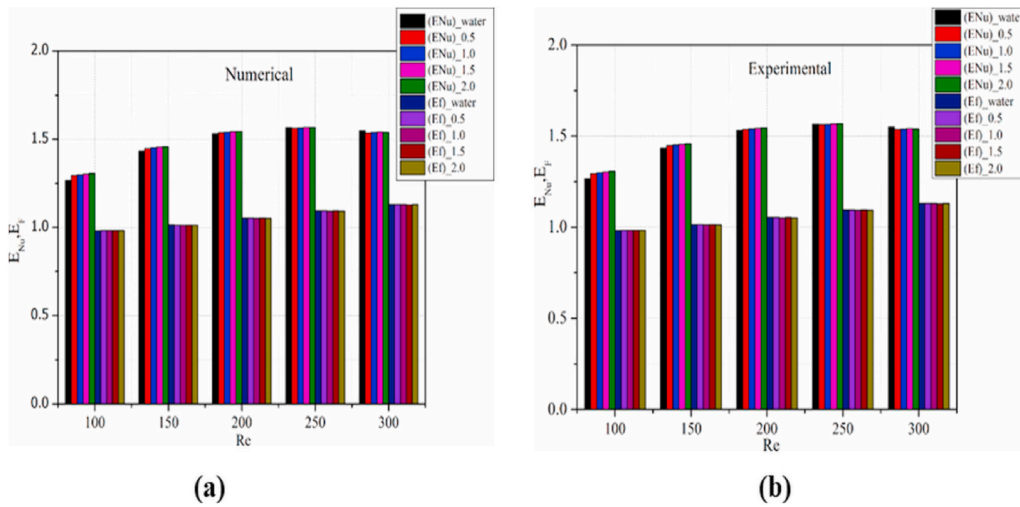


Fig. 23. E_{Nu} , E_f vs. Re (a) numerical (b) experimental.

Fig. 23 compares the E_{Nu} of OFHS at the $Re = 100$ to 300 . As the Re increases from 100 to 300 , the higher heat transfer augmentation occurs with slightly increasing E_f , leading to somewhat higher pumping or fanning power even though it is manageable by the same gear pump. Therefore, an increased heat transfer rate for improved geometry cannot explain the actual scene of enhanced performance. Thus, the heat sink geometry has been meticulously constructed to keep the friction factor as low as possible. An OFHS design is achieved for improved heat transfer. The thermal efficiency of the OFHS is investigated to determine whether inducing vortices in the secondary channel flow and/or adding nanoparticles to the working fluid improves it at a given pumping capacity.

The term performance factor (PF), specified Eq. (48), and the detailed derivation is given in the Appendix [51] correlate the two parameters as mentioned earlier simultaneously. It is defined as the ratio of Nusselt number of enhanced heat transfer surface (OFHS) to smooth channel (SCHS) at the identical Reynolds number. It can be obtained by either experimental or via numerical simulations.

$$PF = \frac{Nu_e}{Nu_o} \bigg|_{p_{fan}} = \frac{\frac{Nu_e}{Nu_o} \bigg|_{Re}}{\left(\frac{f_e}{f_o} \bigg|_{Re} \right)^{\frac{1}{3}}} \quad (48)$$

It is seen from Fig. 24 that the experimental outcomes are well predicted based on numerical results. The performance factor increases with an increase in nanoparticle concentration, and this is because the percentage heat transfer enhancement is higher than the percentage increase in pressure penalty.

6.5. Experimental junction temperature investigation

The average convection coefficient and friction factors are the most widely used parameters to benchmark the thermofluidic characteristics of heat sinks. In other words, the junction temperature is the most critical performance parameter for cooling electronic devices from the standpoint of commercial product design. Irrespective of the type of coolant or the amount of heat dissipation, these limitations must be met healthily and reliably to ensure commercial success. Fig. 25 shows the junction temperatures of OFHS and SCHS concerning Reynolds number. The oblique fin heat sink improves the average junction temperature (T_j) considerably compared to the straight channel heat sink. In addition, the inclusion of nanofluids reduces the experimental junction temperature by up to 6°C in the present experimental investigation. This temperature improvement are of greater significance because the benefits of

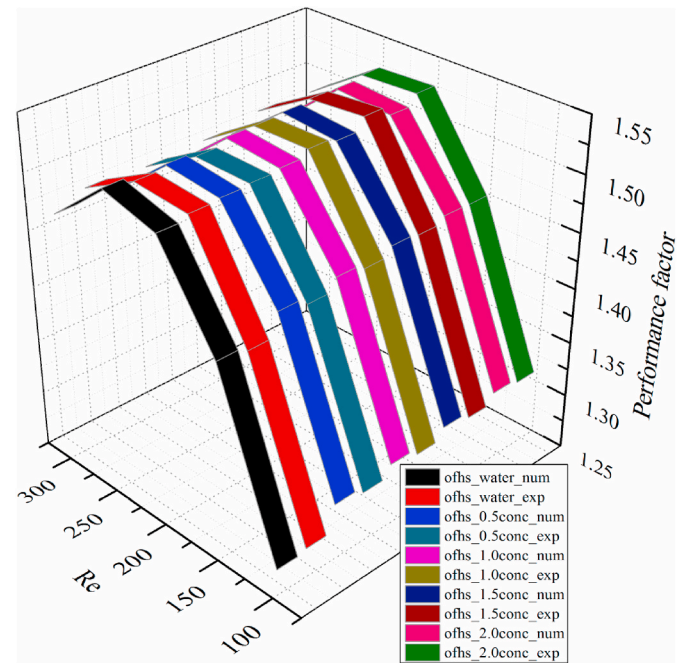


Fig. 24. Performance factor assessment for HSs.

reinitialization and flow mixing at these nanofluid flow rates are more pronounced.

7. Concluding remarks

In this work, an oblique fin heat sink microchannel (OFHS MC) is proposed, and a comprehensive numerical and experimental investigation of the flow and heat transfer is performed systematically. Experiments were conducted for $Re = 100$ – 300 . The volumetric concentrations of Al_2O_3 -water nanofluids are 0.5 , 1.0 , 1.5 and 2% . For comparison, corresponding experiments in SCHS have also been performed. The thermal performance and fluid flow behaviour were investigated using single and full-domain numerical simulations. After the numerical results have been accepted, a detailed analysis of the flow field and temperature field is carried out. After that, both numerically and experimentally, the heat transfer coefficient (h_{avg}), performance factor (PF), and fanning friction factor (f_f) are determined. As a result, all

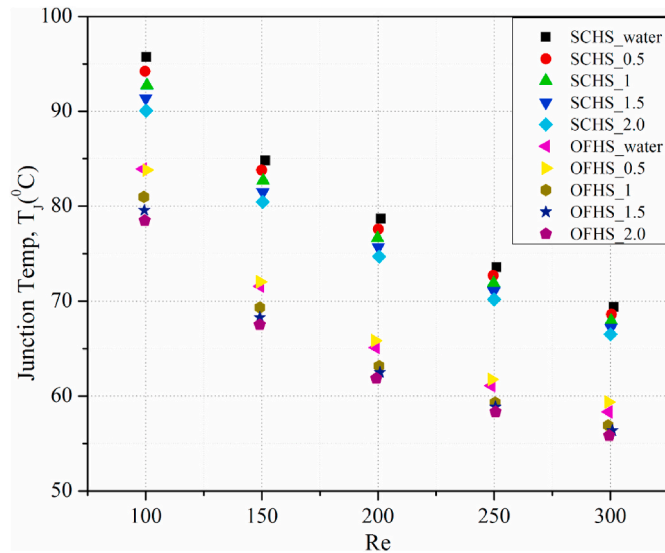


Fig. 25. Junction temperature Vs. Re

results are evaluated, compared, and checked. The significant findings are as follows:

- Boundary layer disruption and vortices generation in the secondary channel is the critical phenomena of improved heat transfer.
- The oblique fin heat sink allows more uniform fluid temperature buildup in the streamwise direction by inducing more intense flow mixing due to the higher secondary nanofluid flow rates.

Nomenclature

MC	Microchannel
OFHS	Oblique fin heat sink
SCHS	Straight channel heat sink
$k-\epsilon$	Turbulence kinetic energy-rate of dissipation of turbulence kinetic energy
W	Width (mm)
PBC	Periodic boundary condition
H	Height (mm)
D_h	Hydraulic diameter (mm)
T	Temperature (K)
k	Thermal conductivity (W/m-K)
C_p	Specific heat (kJ/kgK)
V	Channel Velocity (m/s)
A	Area (m^2)
$\dot{m}$	Mass flow rate (kg/s)
Q	Thermal power (W)
Q_{hb}	Power applied in simulations (W)
Nu	Nusselt number
h	Heat transfer coefficient (W/m ² K)
f	Fanning Friction Factor
P	Pressure (Pa)
L	Length (mm)
Q_{loss}	Heat loss from the test section (W)
$Q_{w/nf, effective}$	Effective heating power (W)
E_{Nu}	Nusselt number Enhancement factor
E_f	Pressure drop penalty

Greek notations

θ	Oblique angle (degree)
ϕ	Volume fraction or volumetric concentration
μ	Viscosity (N.s/m ²)
ρ	Density(kg/m ³)

- The performance factor increases with an increase in nanoparticle concentration, and this is because the percentage heat transfer enhancement is higher than the percentage increase in pressure penalty.
- For the OFHS MC, an enhancement of about 55 % of the heat transfer coefficient with a 2 % volumetric concentration of nanofluid is achieved at $Re = 300$ compared to the SCHS MC at similar conditions.
- The oblique fin heat sink improves the junction temperature considerably compared to the straight channel heat sink. In addition, the inclusion of nanofluids reduces the experimental junction temperature by up to 6 °C.

Funding

The Department of Science and Technology (SERB), India, supported this work (Grant No: ECR/2016/000176). The authors also thank IIT (ISM) Dhanbad, India, for the infrastructural Support extended during the work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Δ Difference

Subscript

ch Channel
 Wet/wetted Wetted wall
 j Junction
 eff effective
 f Base fluid
 p Nanoparticle
 nf Nanofluid
 avg Average
 w/nf Water/nanofluid
 crsec Cross-section
 hb Heat sink base
 m mean
 HS/hs Heat Sink
 amb Ambient
 e Enhanced heat sink (OFHS)
 o reference heat sink design (SCHS)

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